Chapter

5

INFORMATION PROCESSING

I3H7D4

Learning Objectives

- To identify the relationship between the two variables in a pattern.
- To generalize the pattern using tabular form of representation.
- To familiarise the type of patterns in Pascal's triangle.

5.1 Introduction

We can observe patterns in both nature and man made things. In nature, we can see patterns in leaves, trees, snowflakes, movements in celestial bodies and many others. In man made things, we see patterns in structures, buildings, fabric designs, tessellations and many others. This kind of repeated patterns brings happiness as well as interest in knowing more about them. It is really wonder to know all beautiful patterns are based on mathematics. Let us try to find the relationship among the pattern of shapes by representing it in the tabular form in the situations given below.

MATHEMATICS ALIVE - Information Processing in Real Life



Six flags pattern in snowflakes



Patterns in fabric designs

5.2 Tables and Patterns Leading to Linear Functions

Situation 1

Observe the following pattern

Consider a circular disc  . Circular rings of same width are drawn around the disc  . Proceed in the same way to get the shapes following a pattern as shown below.

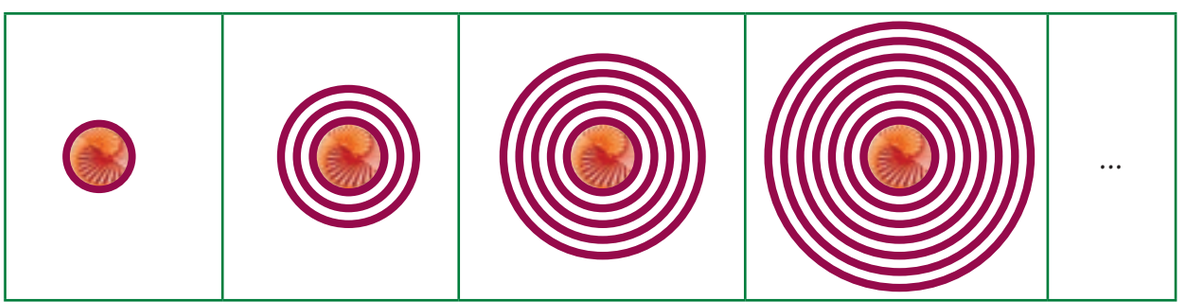


Fig. 5.1